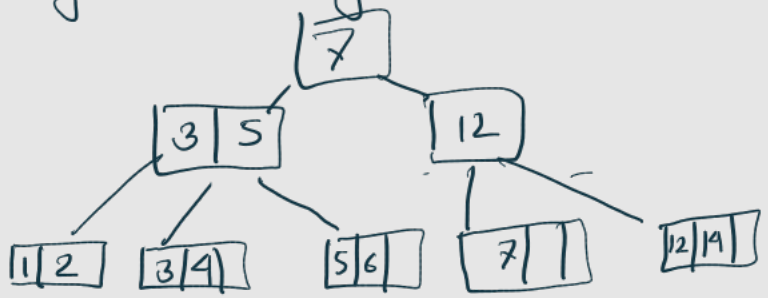
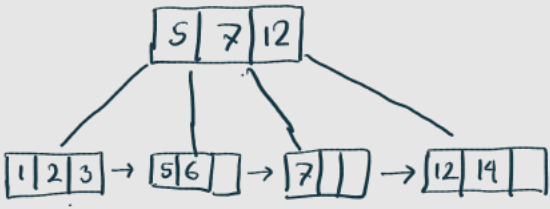


B+ Tree

Condition:

- (1) Keys should be in the min to max range
- (2) LHS and RHS check

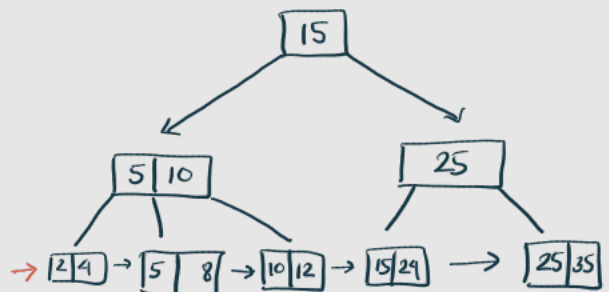
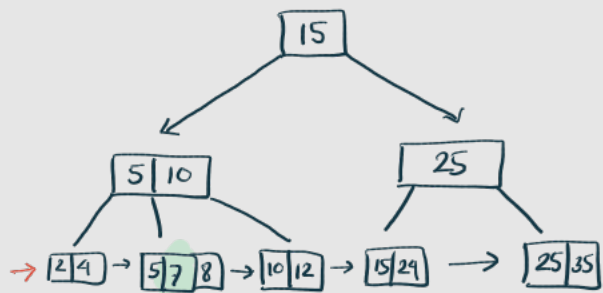
Insert 4 in this B+ Tree using left biasing



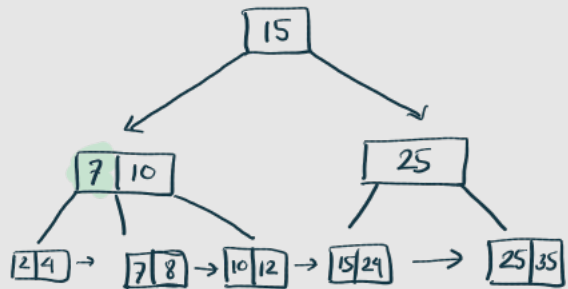
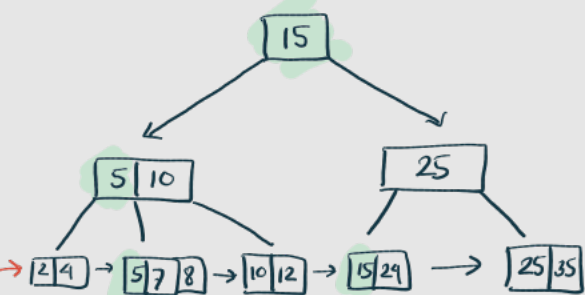
Deletion

$n=4$
 key pointer ^{sub} min ^{max} add
 1 2 3 4

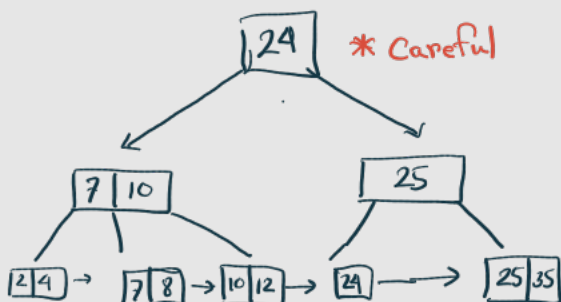
Case 1: For leaf, no of key is greater than minimum key
 1(a) * Delete 7 [$n=4$]



* Delete 5 and 15



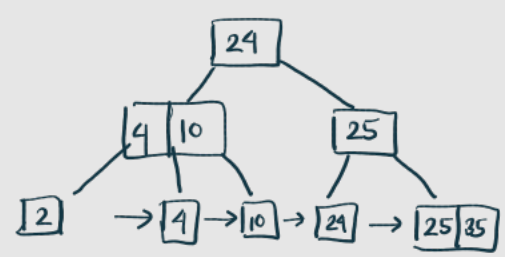
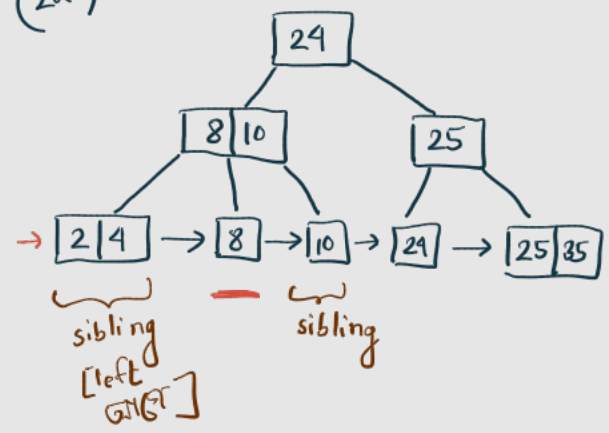
Delete 5



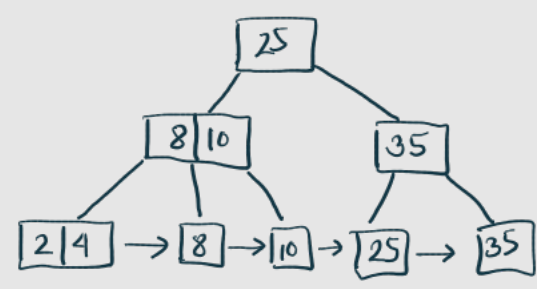
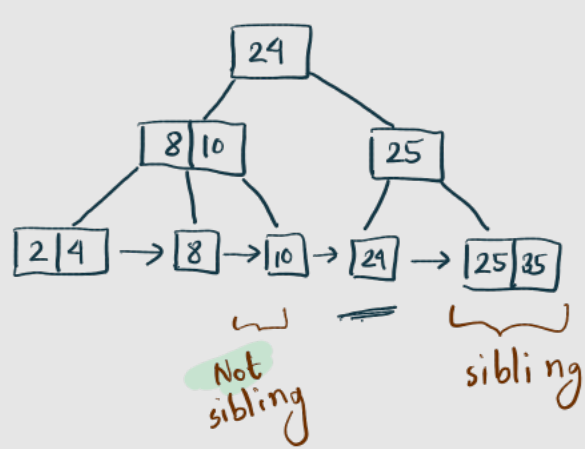
Case 2: Leaf has ^{equal} minimum no of key

(2a)

*Delete 8

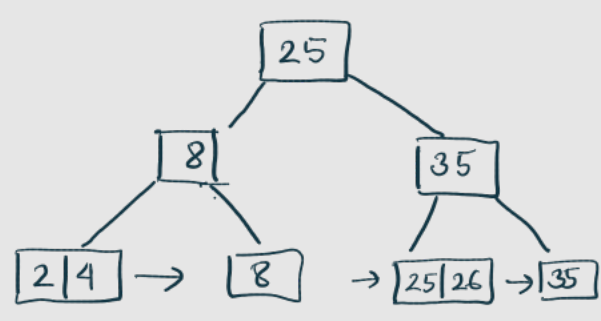
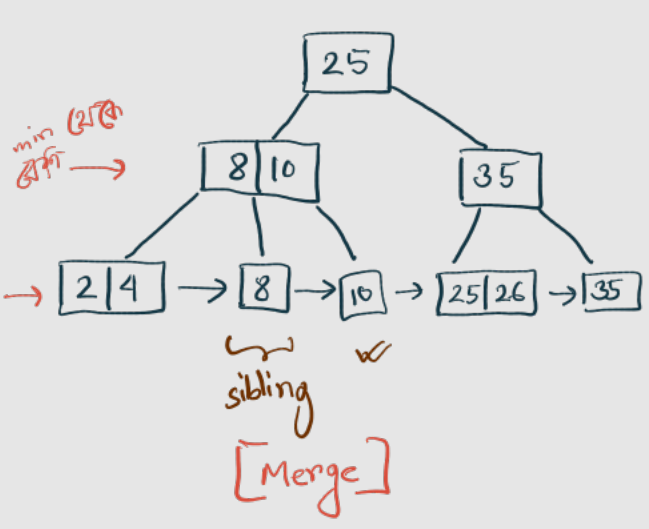


#Delete 24



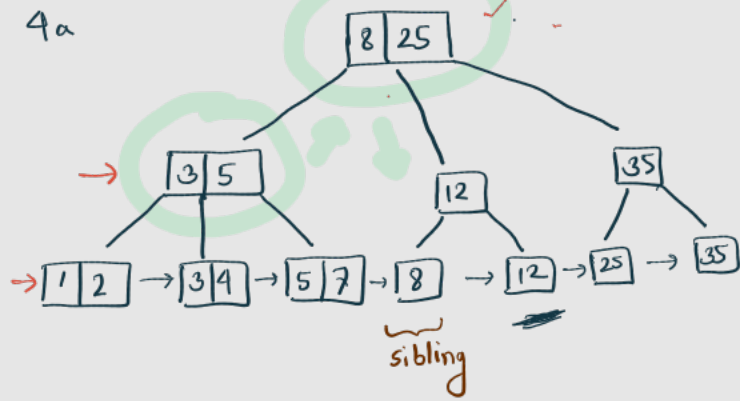
Case 3: Equal to minimum and no borrowing from sibling.

*Delete 10

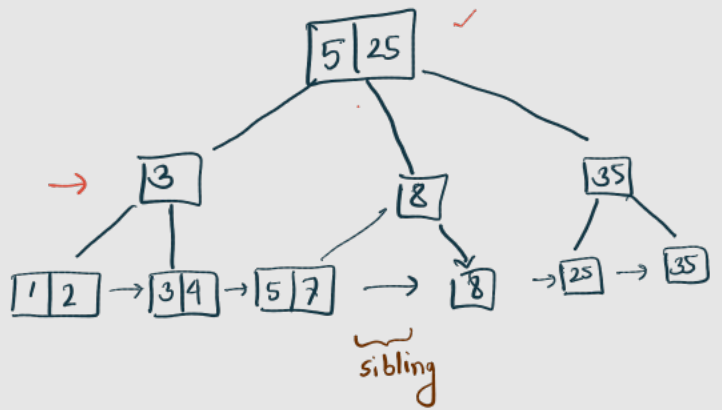


Case 4 :

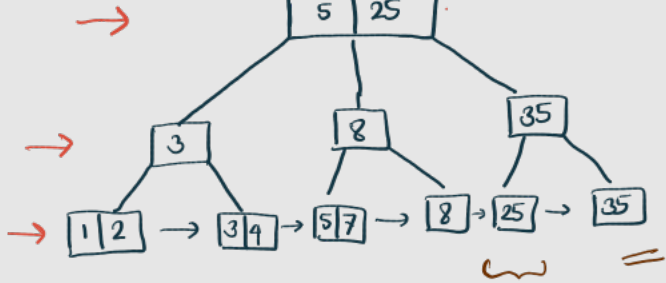
4a



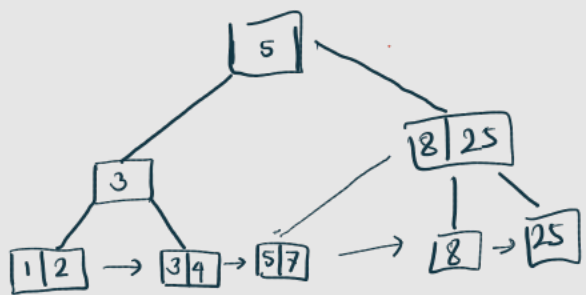
*Delete 12



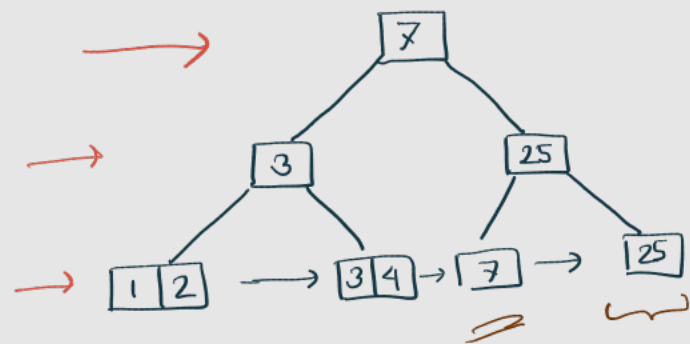
4b



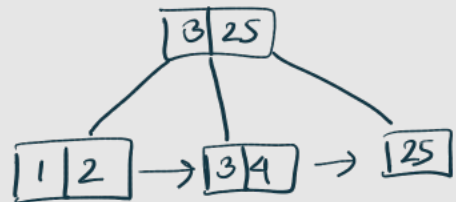
Delete 35



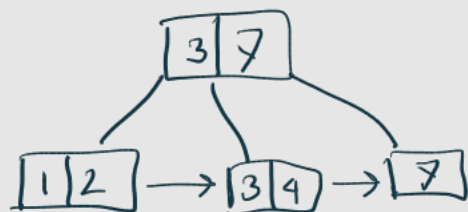
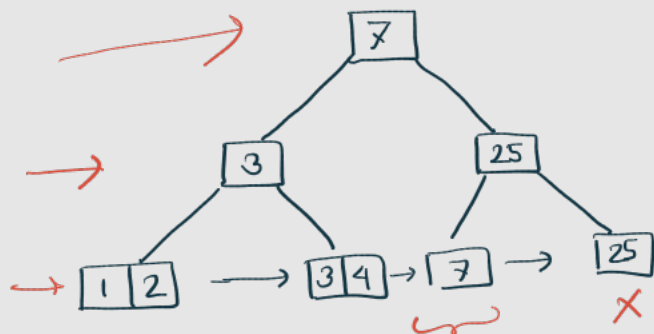
Case 5 :



*Delete 7



#Delete 25



Case 1: >Min

Case 2: =Min ; sibling

Case 3: " , parent

Case 4: " , GP

Case 5: " , team up

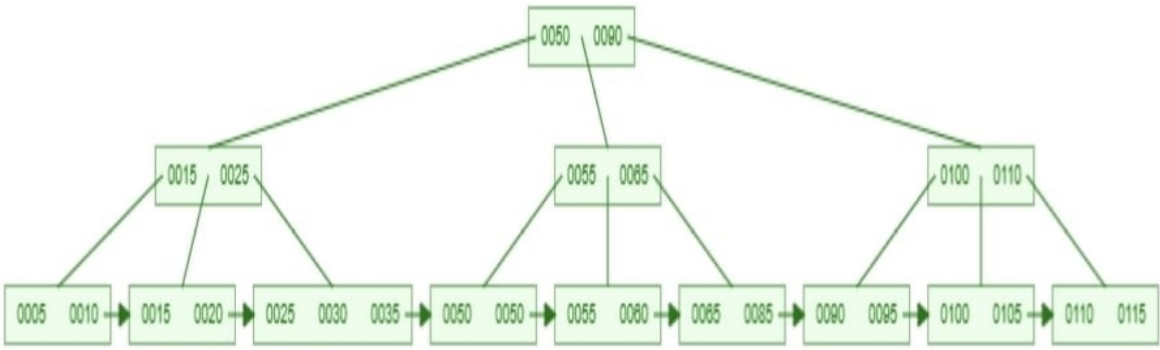
Question 8:

Given the B+ tree below for $n=4$, delete the following values in the given sequence. After each deletion show the updated tree.

35, 90, 95, 55, 25, 30, 20, 100, 50, 50, 65, 60, ~~100~~¹¹⁰, 115, 85

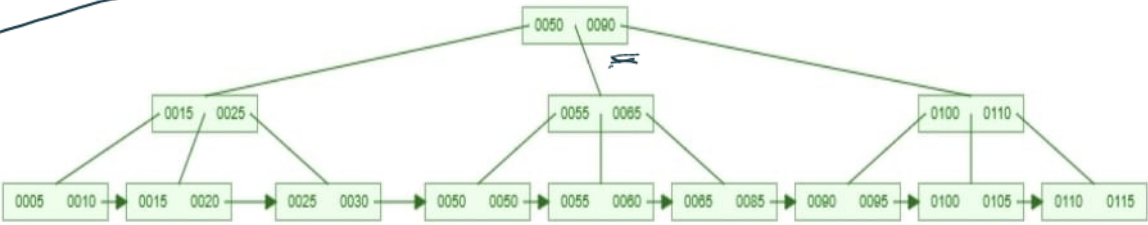
key
points

min max
1 3
2 4

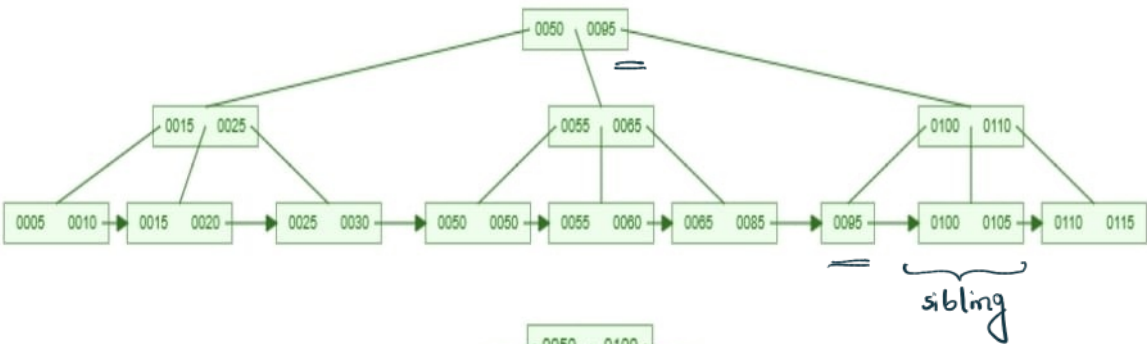


Ans.

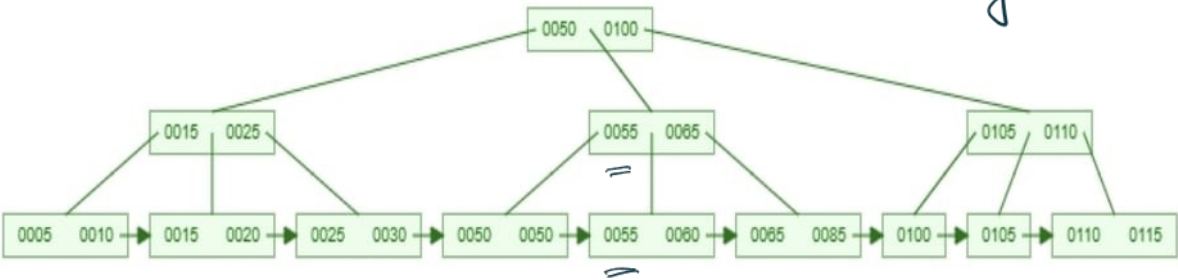
Case 1a



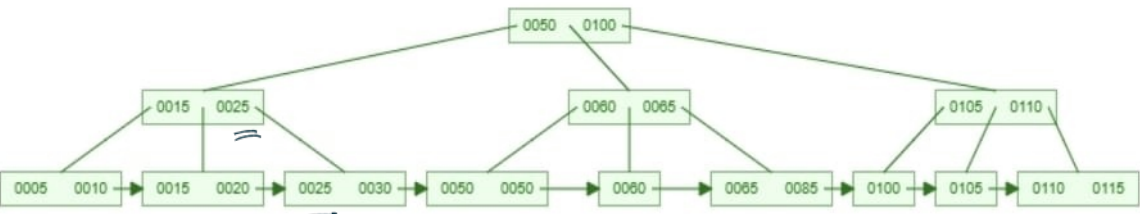
Case 1b



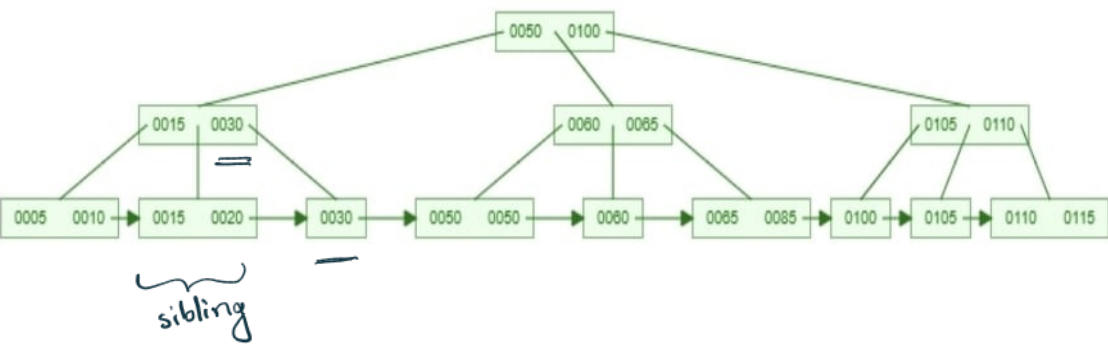
Case 2



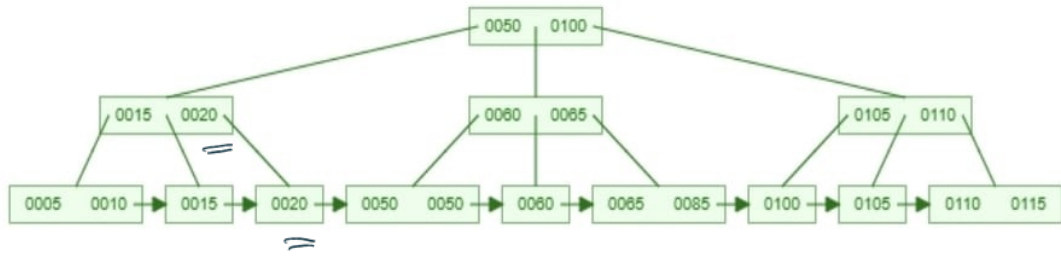
Case 1b



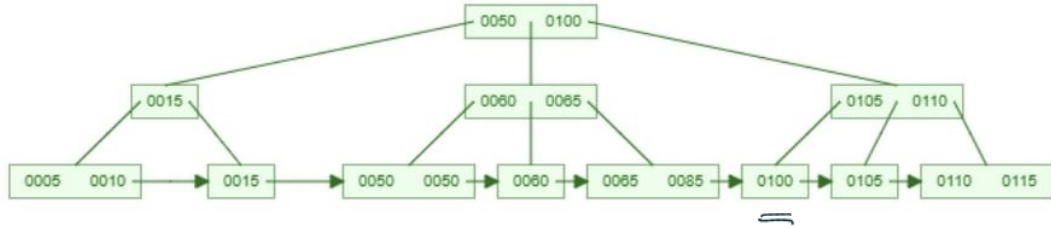
Case 1b



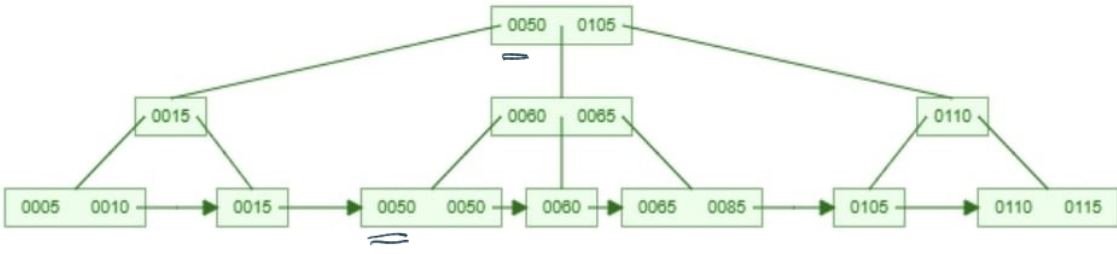
Case 1b



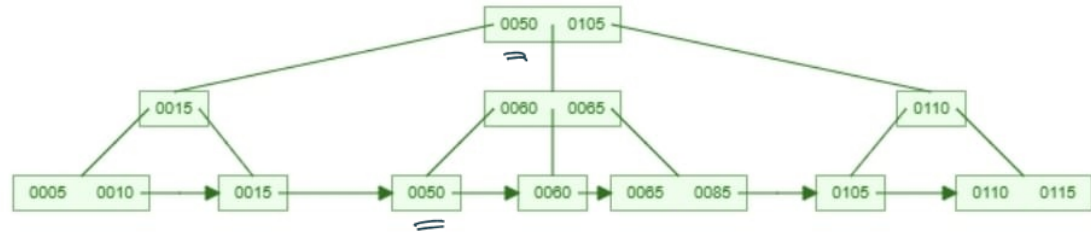
Case 3



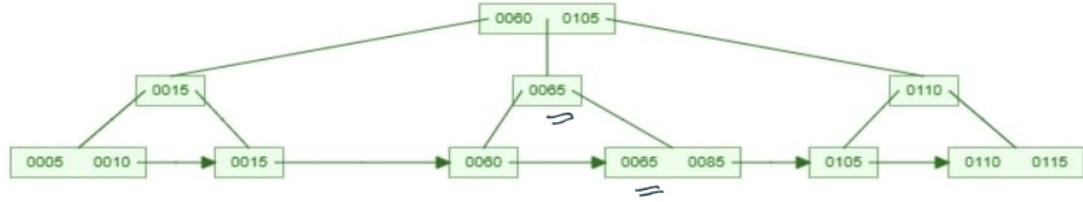
Case 3



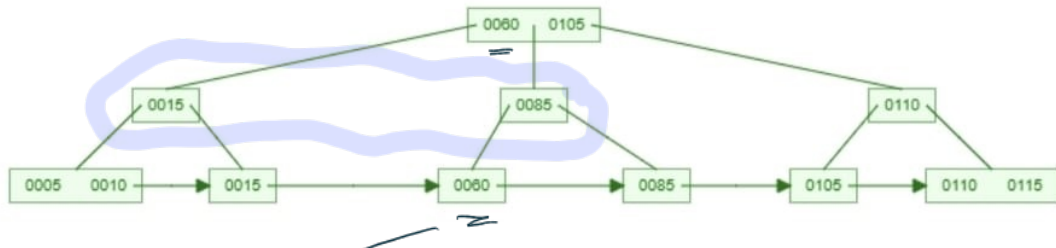
Case 1a



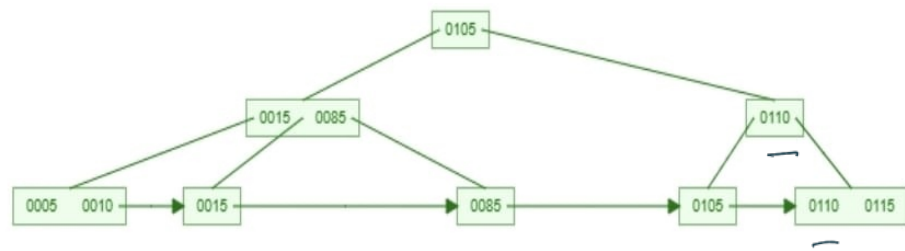
Case 3



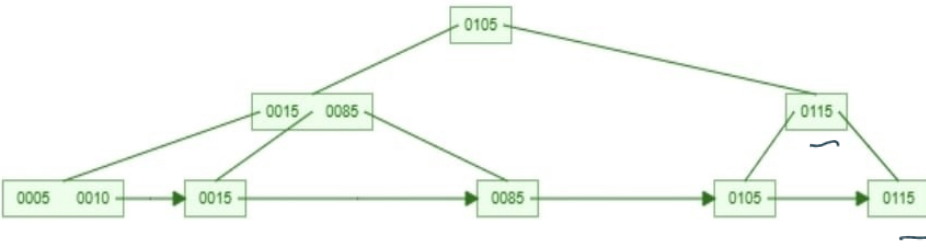
Case 1b



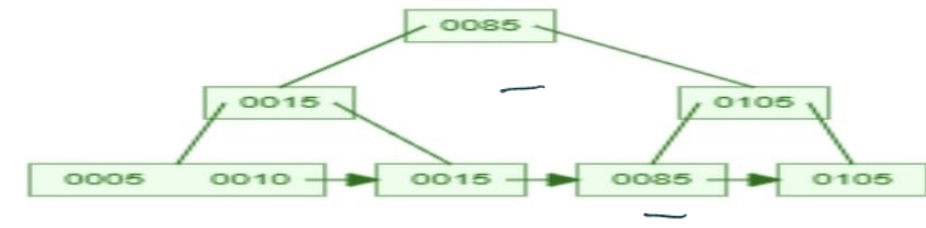
Case 4b



Case 1b



Case 4a



Case 5

